Netgroup Thesis Proposal

Date: February 17, 2026

Iterative approach for Network Security Automation

State-of-the-art solutions (i.e., VEREFOO) for network security automation rely on formal modeling and SMT solvers to automatically configure distributed firewalls in an optimal, formally correct manner, based on a set of Network Security Requirements (NSRs). However, the direct resolution of a monolithic problem does not scale well with network size or the number of security requirements to be enforced.

This thesis aims to repurpose the "React" version of VEREFOO, a custom version that provides additional support for reconfiguring packet-filtering firewalls to satisfy an updated set of NSRs with minimal configuration changes. Specifically, the objective is to further improve scalability by designing an iterative, incremental approach to firewall configuration that reduces solver complexity and exploits the optimizations introduced in React-VEREFOO.

The first goal is to investigate iterative strategies for handling the firewall configuration problem. Instead of solving a single MaxSMT problem over the full set of NSRs, the requirements may be processed one at a time, adding iteratively to the final set of configured NSRs. In this phase, optimization strategies should be explored, such as partitioning NSRs into independent subsets (e.g., based on traffic flows) for incremental configuration. The second goal is to implement the proposed iterative approach using VEREFOO. This includes adapting the current implementation to support NSR partitioning, handle iterative solving, and maintain consistency across iterations. The third and final goal is extensive experimental validation. The iterative approach will be compared with the original VEREFOO (monolithic configuration) to measure computation time, scalability, and the potential loss of global optimality in terms of allocated firewalls and generated rules. Tests will be conducted on realistic and synthetic networks of increasing size and with varying proportions of modified NSRs.

Resources

The following resources are recommended **as a starting point** and should be further explored during the research:

- scientific papers regarding the used network automation tool - VEREFOO - such as: `https://ieeexplore.ieee.org/abstract/document/9737389` (**pdf provided upon request**), focusing mainly on the approach section, and GitHub repo: `https://github.com/netgroup-polito/verefoo`);
- the scientific paper presenting the React-VEREFOO approach `https://ieeexplore.ieee.org/document/10575212` (**pdf provided upon request**).